وزارة الصحة العراقية
IRAQI MINISTRY OF HEALTH
FOUNDED 1920 تأسست

دائرة الصحة العامة
Public Health Directorate

PRIMARY HEARING CARE AT PRIMARY HEALTH CARE CENTERS



2025

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IRAQI MINISTRY OF HEALTH
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Public Health Directorate

(National Guideline for Primary Health Care Physicians)

Primary Hearing Care at Primary Health Care Centers

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Contents:

Introduction……..……1

Mechanism of Hearing and Anatomy of The Ear…….……..…..4

Causes of Hearing Loss and Deafness…….....…….……...……..11

Early Detection of Hearing Loss in New-Born & Children Under Five Years……..…18

Primary Hearing Care for Students …….....……..….29

Hearing Tests at Primary Health Care Centers……33

Common Ear Infections……..…… 39

Ototoxic Medication……...…….……52

Noise Induced Hearing loss……...…...……..…….58

References …….……..……66

Introduction



INTRODUCTION

According to WHO statistics (2024), more than 5% of the world's population- or about430 million people around the world - have disabling hearing loss and 34 million of these are children.

It is estimated that by 2050 over 700 million people will have disabling hearing loss. Hearing loss may result from genetic causes, complications at birth, certain infectious diseases, chronic ear infections, the use of particular drugs, exposure to excessive noise, and ageing.

About 60% of childhood hearing loss is due to preventable causes.

About 1.1 billion young people (aged between 12–35 years) are at risk of hearing loss due to exposure to noise in recreational settings.

Unaddressed hearing loss poses an annual global cost of US$ 980 billion. Interventions to prevent, identify and address hearing loss are cost-effective and can bring great benefit to individuals.

People with hearing loss benefit from early identification; use of hearing aids, cochlear implants and other assistive devices; captioning and sign language; and other forms of educational and social support.

More than one fourth of people over 60 years of age are affected by disabling hearing loss.

Almost 80% of people suffering from disabling hearing loss are in low- and middle-income countries.

HEARING IMPORTANCE:

1. Improve communication skills with others.
2. Education.
3. Development of speech and language skills.

Hearing loss:

A person who is not able to hear as well as someone with normal hearing – or has hearing thresholds of 20 dB or better in both ears – is said to have **hearing loss.**

- **Disabling hearing loss** refers to hearing loss greater than 35 decibels (dB) in the better hearing ear in adults and a hearing loss greater than 30 dB in the better hearing ear in children. The majority of people with disabling hearing loss live in low- and middle-income countries.

IMPACT OF HEARING LOSS:

Functional impact:

One of the main impacts of hearing loss is on the individual's ability to communicate with others. Spoken language development is often delayed in children with unaddressed hearing loss.
Unaddressed hearing loss and ear diseases such as otitis media can have a significantly adverse effect on the academic performance of children. They often have increased rates of grade failure and greater need for education assistance.

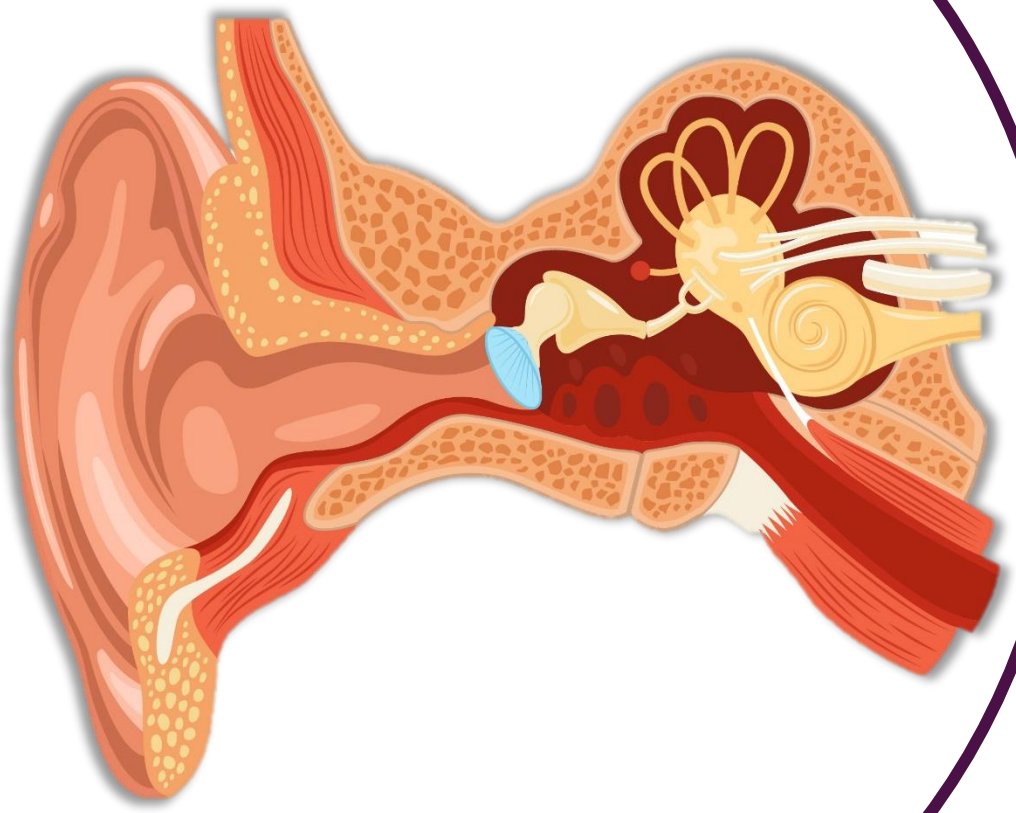
Social and emotional impact:

Exclusion from communication can have a significant impact on everyday life, causing feelings of loneliness, isolation, and frustration, particularly among older people with hearing loss.

Economic Impact:

WHO estimates that unaddressed hearing loss poses an annual global cost of US$ 980 billion. This includes health sector costs (excluding the cost of hearing devices), costs of educational support, loss of productivity, and societal costs.
In developing countries, children with hearing loss and deafness rarely receive any schooling. Adults with hearing loss also have a much higher unemployment rate. Among those who are employed, a higher percentage of people with hearing loss are in the lower grades of employment compared with the general workforce.

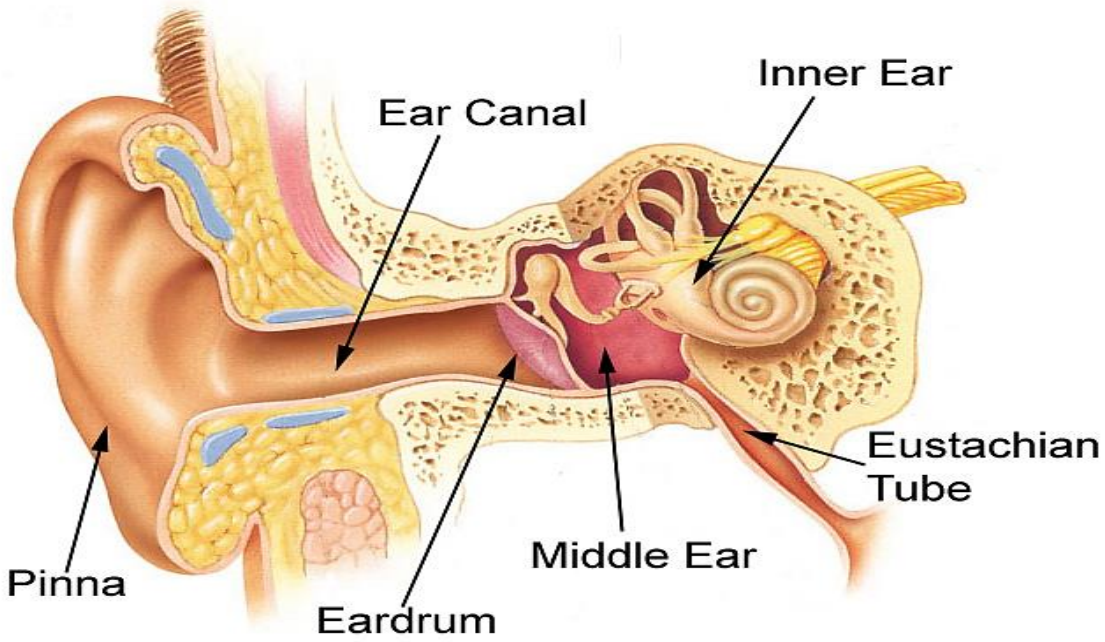
Mechanism of Hearing and Anatomy of The Ear



PRE-TEST

Questions	True	False	Don't know
The pinna is made up of cartilage covered by skin and can bend			
The tragus makes wax to fill up the ear canal			
The normal ear canal is a straight tube with a wet lining like the lining inside your mouth			
The eardrum is at the end of the ear canal			
The ossicles conduct sound vibrations from the eardrum to the cochlea (hearing organ)			
The Eustachian tube connects the middle ear space to the back of the nose			
The middle ear space is full of mucus to help us hear properly			
The cochlea (hearing organ) contains hair cells to convert sound vibrations into nerve signals			
The balance organ is called the vestibular system			
Infection in the ear can cause a lame face (facial palsy)			

STRUCTURE OF THE EAR



❖ THE OUTER /EXTERNAL EAR:

The outer ear has two parts – the pinna and the ear canal

- **The structure of the pinna:**
  The pinna is made out of cartilage covered by normal skin. Cartilage can be bent. The ear lobe is soft and this is where holes are usually made so that earrings or traditional jewelry can be worn. The pinna leads into the ear canal. In front of the opening of the ear canal is a bump called the tragus.

- **The function of the pinna:**
  The pinna picks up sound vibrations and directs them into the ear canal. Hard of hearing people sometimes put their hand behind the pinna to help pick up more sound.
  If a person has deformed ears they might have difficulty hearing. Some deformed ears have no opening into the ear canal so that sounds cannot get through which will make the hearing problem much worse.

- **The structure of the ear canal:**
  The ear canal is an open tube with a skin lining. At the entrance there are hairs to try and stopthings getting into the ear canal. Just beyond the hairs are glands that produce wax

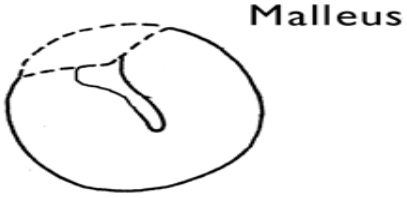
that spreads tocover the skin in the ear canal and helps to keep it healthy. The ear canal normally cleans itself andclears the wax out by itself.
There is a small bend in the ear canal and in the deep part the skin is thin and sensitive to pain and can easily be injured. At the end of the ear canal is the eardrum.

- **The function of the ear canal:**
  The sound vibrations picked up by the pinna travel down the ear canal to the eardrum and makethe eardrum vibrate.
  If the ear canal is blocked the sound vibrations cannot reach the eardrum. When this happens theperson is not able to hear sounds clearly. The commonest cause of a blocked ear canal is wax.
  Some people produce more wax than normal and some ears do not clear the wax out properly.

❖ THE MIDDLE EAR:

- **The structure of the middle ear:**
  The ear canal leads to the eardrum. The eardrum is a thin membrane that separates the outer ear from the middle ear. The middle ear is a space that is filled with air. The air gets into the middle ear through the Eustachian tube. The Eustachian tube goes from the back of the nose to the middle ear.
  There are three tiny bones in the middle ear called the ossicles
  Ossicles = malleus + incus + stapes
  The malleus forms part of the eardrum. The malleus is connected to the incus. The incus is connected to the stapes. The stapes fits into a tiny oval window that opens into the inner ear.
  The middle ear has a lining that usually secretes a tiny amount of mucus. This mucus is drained away through the Eustachian tube. If the lining produces too much mucus this block up the Eustachian tube and air cannot get into the middle ear and the mucus cannot drain away properly. The middle ear space will fill up with mucus and then the eardrum and the ossicles will not be able to vibrate properly to transmit sound. The patient may not hear well and have some pain. Bacteria can get through the Eustachian tube into the middle ear and cause infection.





- **The function of the middle ear:**
  The eardrum picks up the sound vibrations that travel down the ear canal. The eardrum vibrates and makes the ossicles vibrate. The ossicles vibrate and conduct these vibrations across the middle ear space. The stapes vibrates and make the fluid in the inner ear vibrate. The sound vibrations can only be conducted across the middle ear if the space is filled with air.

- **The mastoid air cell system:**
  The mastoid bone is situated behind the middle ear and is filled with air.
  The mastoid is a bony system of cells that contain air and is linked to the middle ear. Both these areas are filled with air and the mastoid air cell system works as an air tank that helps balance the air pressure changes in the middle ear.

❖ THE INNER EAR:

The inner ear has two parts.
- The cochlea deals with sound vibrations and is responsible for hearing
- The vestibular system is responsible for balance

- **Hearing: the cochlea:**
  The cochlea is filled with fluid and contains a delicate membrane lined with tiny hair cells. The hair cells are all connected to the hearing nerve. Vibrations of the ossicles make the fluid vibrate. The vibrations are picked up by the hair cells. The hair cells change the sound vibrations into tiny nerve signals. These nerve signals then travel along the auditory nerve to the brain. In the brain there is a special area where these nerve signals are interpreted as the sounds we hear. Damage to the cochlea or hearing nerve causes hearing impairment or deafness.

The path does sound travel from outside the ear until we can hear it:
Sound vibrations are picked up by the pinna
The pinna directs these sound vibrations into the ear canal
The sound vibrations reach the membrane of the eardrum and make it vibrate
The vibration of the eardrum causes the ossicles to vibrate
The ossicles vibrate and cause the fluid in the inner ear to vibrate
These vibrations are picked up by the hair cells in the cochlea
These hair cells change the vibrations into nerve signals
These nerve signals travel along the hearing nerve to the brain
In the brain the nerve signals are interpreted into the sounds we hear.

- **Balance: The vestibular system:**
  The vestibular system is also filled with fluid. Movements of our head cause the fluid to move.
  There are different hair cells in the vestibular system that is all connected to the balance nerve.
  These hair cells pick up any movement in the fluid and convert it into nerve signals. These nerve signals pass along the balance nerve into the brain. The brain has a special area where the nerve signals are interpreted as the movements we feel.
  Normal function on both sides is needed for us to keep our balance and so that we can keep our eyes focused on what we are looking at as we move our head around.
  If there is infection in one ear that spreads into the inner ear, the function of the vestibular system on that side will be upset and the patient will feel dizzy. If there is inflammation of the balance nerve on one side the same thing will happen. When a patient is dizzy and feels as if everything is moving round and round we say that they have vertigo.

➢ FACIAL NERVE:

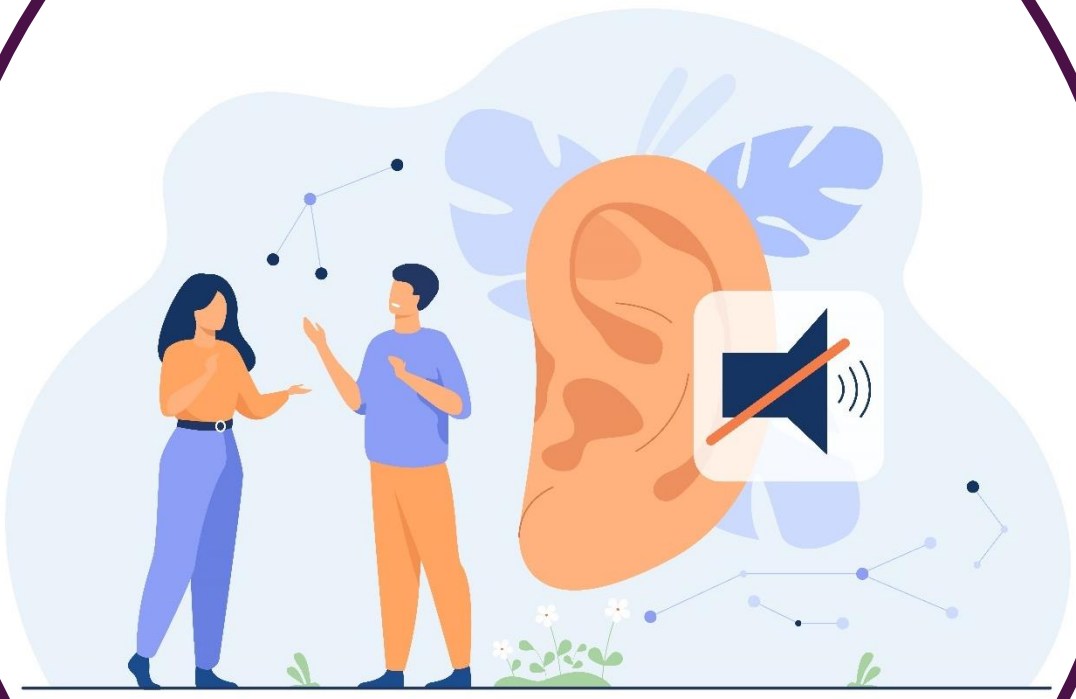
In the face there is only one nerve on each side that makes all the muscles on that side of the face work. It is called the facial nerve. Infection in the middle ear can cause facial palsy (also called lame face).
The facial nerve goes through the ear inside a bony tunnel on the wall between the middle ear and the inner ear.



POST-TEST

Questions	True	False	Don't know
The pinna is made up of cartilage covered by skin and can bend			
The tragus makes wax to fill up the ear canal			
The normal ear canal is a straight tube with a wet lining like the lining inside your mouth			
The eardrum is at the end of the ear canal			
The ossicles conduct sound vibrations from the eardrum to the cochlea (hearing organ)			
The Eustachian tube connects the middle ear space to the back of the nose			
The middle ear space is full of mucus to help us hear properly			
The cochlea (hearing organ) contains hair cells to convert sound vibrations into nerve signals			
The balance organ is called the vestibular system			
Infection in the ear can cause a lame face (facial palsy)			

Causes of Hearing Loss and Deafness



PRE-TEST

Questions	True	False	Don't know
Babies born with low birth weight are at risk for hearing loss			
Certain medicines can cause damage to hearing			
Damage to the auditory nerve causes conductive hearing loss			
A person with normal hearing has a hearing threshold of less than 20 decibels in the better hearing ear			
Vaccination can prevent hearing loss			
60% of hearing loss in children is due to preventable causes			

CAUSES OF HEARING LOSS AND DEAFNESS

The causes of hearing loss and deafness can be congenital or acquired.



1. Congenital causes:

Congenital causes may lead to hearing loss being present at or acquired soon after birth. Hearing loss can be caused by hereditary and non-hereditary genetic factors or by certain complications during pregnancy and childbirth, including:

- maternal rubella, syphilis or certain other infections during pregnancy;
- low birth weight;
- birth asphyxia (a lack of oxygen at the time of birth)
- inappropriate use of particular drugs during pregnancy, such as amino glycosides, cytotoxic drugs, antimalarial drugs, and diuretics;
- Severe jaundice in the neonatal period, which can damage the hearing nerve in a newborn infant.

2. Acquired causes:

Acquired causes may lead to hearing loss at any age, such as:

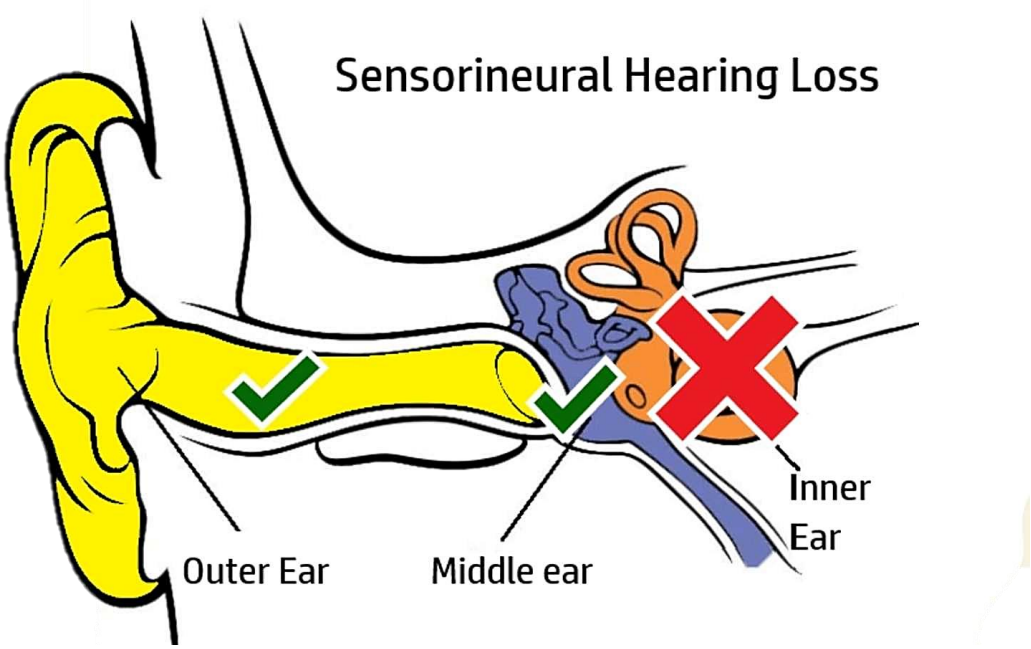
- Infectious diseases including meningitis, measles and mumps.
- Chronic ear infections.
- Collection of fluid in the ear (otitis media).
- Use of certain medicines, such as those used in the treatment of neonatal infections, malaria, drug-resistant tuberculosis, and cancers.
- Injury to the head or the ear.
- Excessive noise, including occupational noise such as that from machinery and explosions.
- Recreational exposure to loud sounds such as that from use of personal audio devices at high volumes and for prolonged periods of time and regular attendance at concerts, nightclubs, bars and sporting events.
- Ageing, in particular due to degeneration of sensory cells.
- Wax or foreign bodies blocking the ear canal.

> **Among children, chronic otitis media is a common cause of hearing loss.**

Types of Hearing Loss:

1. **Conductive Hearing Loss:**
   When the problem causing hearing loss is present in the ear canal or middle ear. It is difficult to conduct the sound to the inner ear. This problem can usually be corrected by treatment or by hearing aids.

2. **Sensory neural hearing loss:**
   When the problem causing the hearing loss is present in the cochlea or in the auditory nerve or both or as a result of any other form of damage to the nerve or brain.



3. **Mixed hearing loss:**
   It is a hearing impairment that combines conductive hearing impairment and sensorineural hearing impairment.

Grades of hearing loss and related hearing experience:

Grade	Hearing threshold in better hearing ear in decibels (dB)	Hearing experience in a quiet environment for most adults	Hearing experience in a noisy environment for most adults
Normal hearing	Less than 20 dB	No problem hearing sounds	No or minimal problem hearing sounds
Mild hearing loss	20 to < 35 dB	Does not have problems hearing conversational speech	May have difficulty hearing conversational speech
Moderate hearing loss	35 to < 50 dB	May have difficulty hearing conversational speech	Difficulty hearing and taking part in conversation
Moderately severe hearing loss	50 to < 65 dB	Difficulty hearing conversational speech; can hear raised voices without difficulty	Difficulty hearing most speech and taking part in conversation
Severe hearing loss	65 to < 80 dB	Does not hear most conversational speech; may have difficulty hearing and understanding raised voices	Extreme difficulty hearing speech and taking part in conversation
Profound hearing loss	80 to < 95 dB	Extreme difficulty hearing raised voices	Conversational speech cannot be heard
Complete or total hearing loss/deafness	95 dB or greater	Cannot hear speech and most environmental sounds	Cannot hear speech and most environmental sounds
Unilateral	< 20 dB in the better ear, 35 dB or greater in the worse ear	May not have problem unless sound is near the poorer hearing ear. May have difficulty in locating sounds	May have difficulty hearing speech and taking part in conversation, and in locating sounds

Prevention:

Overall, it is suggested that half of all cases of hearing loss can be prevented through public health measures.
In children under 15 years of age, 60% of hearing loss is attributable to preventable causes. This figure is higher in low- and middle-income countries (75%) as compared to high-income countries (49%). Overall, preventable causes of childhood hearing loss include:

- Infections such as mumps, measles, rubella, meningitis, cytomegalovirus infections, and chronic otitis media (31%).
- Complications at the time of birth, such as birth asphyxia, low birth weight, prematurity, and jaundice (17%).
- Use of ototoxic medicines in expecting mothers and babies (4%).
- Others (8%)

Some simple strategies for prevention of hearing loss include:

- Immunizing children against childhood diseases, including measles, meningitis, rubella and mumps.
- Immunizing adolescent girls and women of reproductive age against rubella before pregnancy.
- Preventing cytomegalovirus infections in expectant mothers through good hygiene.
- Screening for and treating syphilis and other infections in pregnant women.
- Strengthening maternal and child health programs, including promotion of safe childbirth.
- Following healthy ear care practices.
- Reducing exposure (both occupational and recreational) to loud sounds by raising awareness about the risks; developing and enforcing relevant legislation; and encouraging individuals to use personal protective devices such as earplugs and noise-cancelling earphones and headphones.
- Screening of children for otitis media, followed by appropriate medical or surgical interventions.
- Avoiding the use of particular drugs which may be harmful to hearing, unless prescribed and monitored by a qualified physician.
- Referring infants at high risk, such as those with a family history of deafness or those born with low birth weight, birth asphyxia, jaundice or meningitis, for early assessment of hearing, to ensure prompt diagnosis and appropriate management, as required.

POST-TEST

Questions	True	False	Don't know
Babies born with low birth weight are at risk for hearing loss			
Certain medicines can cause damage to hearing			
Damage to the auditory nerve causes conductive hearing loss			
A person with normal hearing has a hearing threshold of less than 20 decibels in the better hearing ear			
Vaccination can prevent hearing loss			
60% of hearing loss in children is due to preventable causes			

Early detection of hearing loss in new-born & children under five years



PRE-TEST

Questions	True	False	Don't know
The baby cannot hear the mother's voice during late pregnancy			
At birth, the baby's ears will be full of fluid, which may take some time to completely clear			
Singing nursery rhymes or playing music would not help the baby to recognize and learn new sounds			
Automated otoacoustic emission test is a comfortable hearing test to assess baby hearing			
At 10-15 months, the baby follows one-step commands when shown by a gesture			

Introduction:

Undiagnosed or late-diagnosed hearing loss in young children can have substantial negative consequences: not only in terms of its impact on a child's language and communication development, but also on social and emotional development and mental health, family relationships, educational opportunity and achievement and later economic contribution to society.

Many high-income countries have implemented early hearing detection and intervention. There is now compelling evidence from established programmes that when hearing screening in early life is followed up by appropriate assessment, management and support for both child and family, the developmental outcomes for a child with hearing impairment can be radically improved.

Approximately 34 million children in the world have a disabling hearing loss (35dB or greater) and approximately 1 out of every 1,000 babies are born with congenital hearing loss. Most deaf children develop language at a slower rate than their hearing peers and are at higher risk of mental health problems. However, if hearing loss is detected early and these children are then given appropriate support, this need not necessarily be the case. Deafness is not a learning disability and there is no reason why they should achieve any less than their hearing peers.

At birth

Some babies are born with hearing problems. Other children are born with normal hearing and begin to have hearing problems as they grow older.

Your baby has been able to hear clearly from birth. She could even hear your voice and other noises when she was tucked up in your womb (uterus). As your baby grows, she'll use her hearing to take huge amounts of information about the world around her.

When will my baby be able to hear properly?

Although your baby's ears are well-developed at birth, it may take up to six months before the baby can fully hear and understand a range of sounds.

There are two reasons for this:



- Firstly, at birth, your baby's ears will still be full of fluid, which may take some time to completely clear.
- Secondly, the parts of your baby's brain that deal with hearing are still developing. In spite of this, your baby will love to hear the voices of you and your partner from the moment she's born.

How can I encourage my baby's ability to listen?

You can do plenty of things to help your baby recognize and learn new sounds. Try singing nursery rhymes or playing music. Your baby will be delighted by lots of sounds and music, so you don't have to restrict her to children's songs. Share your own favorite music with her. You may notice how your baby seems to enjoy some sounds and music more than others as she begins to develop her own tastes. The sound of wind chimes or a ticking clock may also amuse your baby. Reading to your baby, no matter how young she is, will pay off too. Hearing you talk will help your baby's language skills.
Try varying the pitch of your voice or singing as you read to her to make it more fun. The more you talk and read to her, the more sounds and words she will learn as she gets ready to talk.



Notes

You baby will be able to sleep through the telephone ringing and the dog barking. This is normal, she needs her sleep.
When your baby is awake and alert, she should startle at loud, sudden noises. She will turn to you when she hears your voice, and seem to react normally to sounds around her. Bear in mind if she has a cold or an ear infection, this may temporarily affect her hearing.

Milestones related to hearing:

Birth to three months

- Reacts to loud sounds with startle reflex
- Is soothed and quieted by soft sounds
- Turns head to you when you speak
- Is awakened by loud voices and sounds
- Smiles in response to voices when spoken to
- Seems to know your voice and quiets down if crying
- She'll also respond to familiar noises, such as your voice or a lullaby you play or sing to her often.

Three to six months

By three months, the part of your baby's brain that helps with hearing, language, and smell will be more receptive and active. This part of your baby's brain is called the temporal lobe. When your baby hears your voice, she may look directly at you and gurgle in an attempt to talk back.

Babbling and listening can be hard work for your baby at three months. If she looks the other way or loses concentration while you talk or read to her, it's not necessarily because she can't hear you. She may just have had enough stimulation.

From four months your baby will react excitedly to sounds, and she may smile when she hears your voice. She may start watching your mouth intently when you speak, and try to copy you. She may start to utter consonant sounds such as "m" and "b".

So, the baby will be able to:

- Looks or turns toward a new sound
- Responds to no and changes in tone of voice
- Imitates his/her own voice
- Begins to repeat sounds (such as ooh, aah, and ba-ba)
- Becomes scared by a loud voice or noise

Six to ten months

At six months or seven months your baby will realize where sounds come from, and she'll turn quickly towards new ones. She'll also be able to respond to very quiet noises, as long as she's not distracted.

- Responds to his/her own name, telephone ringing, someone's voice, even when not loud
- Knows words for common things (cup, shoe) and sayings (bye-bye)
- Makes babbling sounds, even when alone
- Starts to respond to requests such as come here
- Looks at things or pictures when someone talks about them

Ten to fifteen months

By the time your baby is a year old, she'll be able to recognize her favorite songs. She may even try to join in!

- Plays with own voice, enjoying the sound and feel of it
- Points to or looks at familiar objects or people when asked to do so
- Imitates simple words and sounds; may use a few single words meaningfully
- Enjoys games like peek-a-boo and pat-a-cake
- Follows one-step commands when shown by a gesture

Fifteen to eighteen months

- Follows simple directions, such as give me the ball without being shown
- Uses words he/she has learned often
- Uses 2 to 3 word sentences to talk about and ask for things
- Knows 10 to 20 words
- Points to some body parts when asked

Eighteen to twenty-four months

- Understands simple yes-no questions (Are you hungry?)
- Understands simple phrases (in the cup, on the table)
- Enjoys being read to
- Points to pictures when asked

Twenty-four to thirty-six months

- Understands not now and no more
- Chooses things by size (big, little)
- Follows two-step commands, such as get your shoes and come here
- Understands many action words (run, jump)
- Has a word for almost everything
- Uses two- or three-word phrases to talk about and ask for thing
- Speaks in a way that is understood by family members and friends
- Names objects to ask for them or to direct attention to them

Three to four Years

Hears you when you call from another room

- Hears the television or radio at the same sound level as other family members
- Answers simple “Who?” “What?” “Where?” and “Why?” questions
- Talks about activities at day care, preschool, or friends’ homes
- Uses sentences with four or more words
- Speaks easily without having to repeat syllables or words
- He answers by; YES, NO

Four to five Years

Pays attention to a short story and answers simple questions about it

- Hears and understands most of what is said at home and in school
- Uses sentences that give many details
- Tells stories that stay on topic
- Communicates easily with other children and adults
- Says most sounds correctly except for a few (l, s, r, v, z, ch, sh, and th)
- Uses rhyming words
- Names some letters and numbers

Distraction test:

Find a quiet room and someone to help you.

It needs two people to test the hearing, one to stand 1 meter behind and to one side of the baby and make a noise with the noise maker and the other sit in front of the baby and watch to see whether the baby turns to look to see where the sound is coming from.

Ask the mother to hold her baby sitting up on her lap and facing in front while the test is being done.

The person in front gets the baby's attention by showing a toy or something else.

The person in front covers the toy and at the same moment the person testing makes a noise with the noisemaker.

Does the baby look to see where the sound is coming from?

If the baby responds then the person standing behind moves to the other side and the test is repeated.

If the baby responds then the hearing is normal. The test should be repeated after three months to see if the hearing is normal.

If the baby does not respond and there is an ear problem the problem should be treated and the test repeated when it has healed.

If the baby does not respond and there is no ear problem then the parents should be given advice about the things they should do at home to help the baby - see below in the section "Advice to parents with a hearing impaired baby or child". The test should be repeated every three months until both the Health Care Worker and the parents are sure that there is hearing impairment. The child should then be referred to hospital or a local hearing testing center, to find out if the child needs a hearing aid, and to a teacher at a local school, local deaf groups or a "Deaf School" to be assessed to find out whether Special Education is needed.

When to suspect hearing problem?

Hearing problems may be suspected in children who are not responding to sounds or who are not developing their language skills appropriately.

The following are some age-related guidelines that may help to decide if your child is having hearing problems. It is important to remember that not every child is the same, and children reach milestones at different ages.

It's important to know what to expect as your baby grows, because hearing problems can delay the development of voice, speech, and language skills. The checklist below presents the average age by which most babies accomplish a variety of early speech and language

skills. Typically, a child may not accomplish all the items in an age category until he or she reaches the upper age in the age range.

When to do the tests?

Your baby will be offered a hearing test shortly after birth, either in hospital, a community clinic, or at home. It's called the automated otoacoustic emission (AOAE) test. It only takes less than 15 minutes and won't be uncomfortable for your baby.

If the test doesn't show a clear result, you may be asked to come back for another test. This doesn't necessarily mean that there's a problem. Your baby may have been unsettled at the time of the test, or there may be fluid or a temporary blockage in her ear. Or there may have been background noise during the screening.

If the test shows that there could be a problem with your baby's hearing, you'll be invited to a follow-up assessment.

If your baby's hearing tests show no problems but you're still worried, asks your doctor or health visitor to check her hearing again.

When to suspect that baby will have hearing problem

It's rare for babies to have hearing problems, but occasionally it can happen, particularly if:

Your baby had to spend time in a neonatal unit after birth

You had rubella, toxoplasmosis or cytomegalovirus during pregnancy

There is a family history of hearing loss or deafness

Your baby was born prematurely or with a low birth weight

Hearing difficulty, it will be detected straight away. The test is simple and quick, and won't hurt or be uncomfortable for your baby.

Some babies with perfect hearing don't seem to notice noises around them. If your baby sleeps through the phone ringing and the dog barking, don't worry. It's unlikely to mean that her hearing is affected. She probably just needs her sleep.

If you want to test your baby's hearing for yourself, the best time to do it is when she's wide awake and alert. Here are a few quick tests you can do:

If you want to test your baby's hearing for yourself, the best time to do it is when she's wide awake and alert. Here are a few quick tests you can do:

If your baby is **under three months,** clap your hands behind her head. If she startles, she's fine. If she doesn't, repeat the claps a few times.

If your baby is **a bit older, between four months and six months,** she should have some head control. Call her name to see if she turns towards you or reacts to your voice. She may also turn her eyes or head to look for an interesting sound.
As hearing and talking are linked, by four months your baby should start to make cooing sounds and other noises.

Between six months and ten months, your baby should respond to her name and familiar sounds, such as the ringing of the phone or the roar of the vacuum cleaner. She may also respond to quiet noises if she's not occupied with other things.

Between ten months and fifteen months, she should be able to point to a familiar object in a picture book when asked, and react when you say her name. She may even notice quieter sounds coming from another room, and may respond to an expression such as "bye bye", even when no gestures are used.
If your baby doesn't respond to any of these tests, or if you're not certain and want a second opinion, talk to your doctor or health visitor. It is likely to be only a temporary loss caused by a cold. Or she may simply be too engrossed in something else to take notice.

What can affect the baby's hearing?
The most common cause of hearing impairment in babies is an ear infection. The medical name for this is otitis media.
An ear infection causes the build-up of fluid in the middle ear. Mucus or fluid fills the space between your baby's ear drum and inner ear. This usually happens when she has a cold, and if it won't go away it causes glue ear.
Usually glue ear clears up by itself. However, your baby's doctor can treat ear infections that won't go away, so any hearing difficulties should not last. Even if your baby has recurring bouts, they shouldn't leave any lasting impairment
Occasionally, babies' hearing is affected by other factors, including:

- Exposure to prolonged, high-decibel noise such as that found near; airport, runways or at rock concerts.
- Serious complications during birth that may have prevented enough oxygen reaching the baby.
- Low Apgar scores (how a baby's health is measured at birth).
- Exposure during pregnancy to infections such as rubella.

- A family history of hearing problems.
- Premature birth or low birth weight.
- A head injury.

POST-TEST

Questions	True	False	Don't know
The baby cannot hear the mother's voice during late pregnancy			
At birth, the baby's ears will be full of fluid, which may take some time to completely clear			
Singing nursery rhymes or playing music would not help the baby to recognize and learn new sounds			
Automated otoacoustic emission test is a comfortable hearing test to assess baby hearing			
At 10-15 months, the baby follows one-step commands when shown by a gesture			

Primary Hearing Care for Students



Primary Hearing Care for Students:

School health services are an important component of primary health care because they deal with a large and important part of the society. The students constitute about one third of the population of the society and deal with a major and essential part of the society. Early detection and treatment of diseases is a major focus on health care is an important pillar that supports its fundamentals in providing affordable, acceptable, scientifically sound, basic and simple services at low cost.

To provide preventive, curative and rehabilitative health services with quality to students, the National School Health Strategy in Iraq (2018-2022) was prepared and launched for 2018-2019. A detailed annual plan is also prepared, which includes primary hearing care for screening new students and other stages in coordination with relevant authorities and disseminate it to health departments to work with.

The annual plan includes strengthening the capacities of school health workers at various levels as follows:

- Implementing central training workshops for the heads of the school health sections in health departments in the governorates and those responsible for the hearing aids for the purpose of early detection of hearing impairment.
- The implementation of peripheral training workshops by the School Health section in cooperation with the centers of hearing and communication clinics and counseling in the health services for training on the measurement of hearing loss.
- Training the school health coordinator on early detection of hearing impairment and referral mechanism.

Provides primary hearing care and early detection of hearing impairment and ear problems integrated into school health services in phc centers.

Treatment services are provided to the students by the medical unit's doctor in the phc centers and by the dentist for oral and dental health services. Cases are referred to other therapeutic institutions such as hospitals, consulting clinics or private committees when needed. Follow-up nutrition is monitored by the health center.

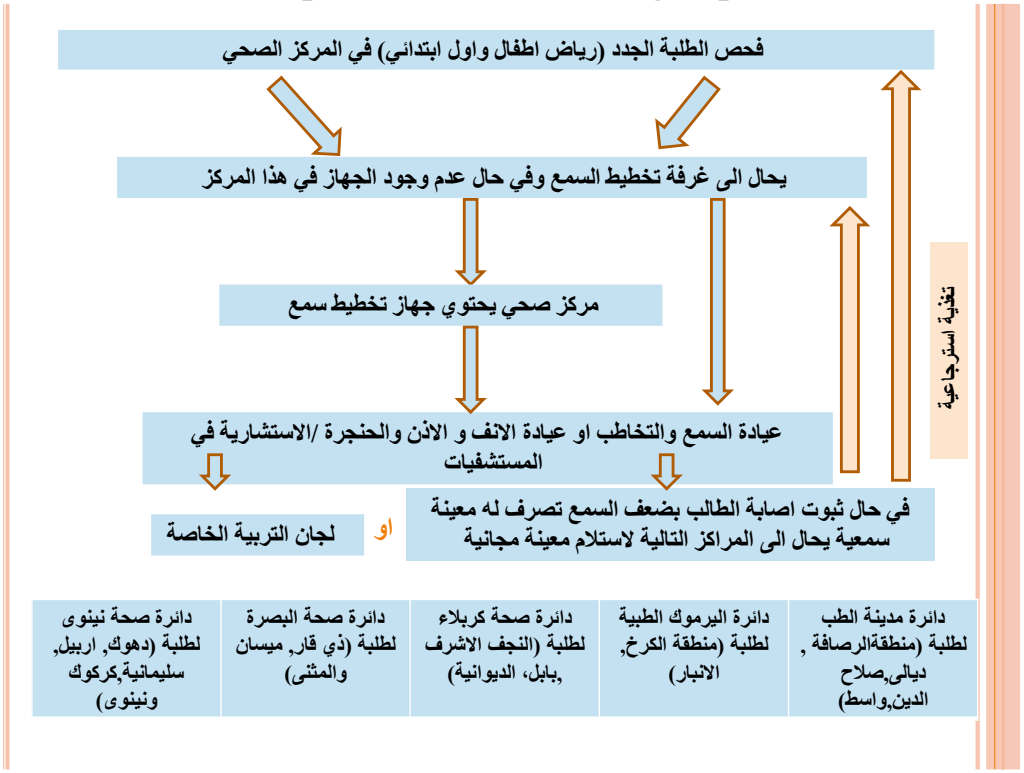
Preventive health services are provided through the school health team during the field visits of schools within the geographical area of each health center to follow the indicators of the school environment and school libraries in cooperation with the health coordinator in the school. Other preventive measures are also implemented, such as early detection of the visually impaired, hearing and other cases.

The rehabilitation services are provided through the workshops for the manufacture of glasses and hearing and communication centers of the health departments in Baghdad and the provinces.
Audiometers are located in specific health centers within each sector in all health departments for the purpose of screening students who are likely to have hearing impairment whether during the examination of new students or referred from schools to the health center.

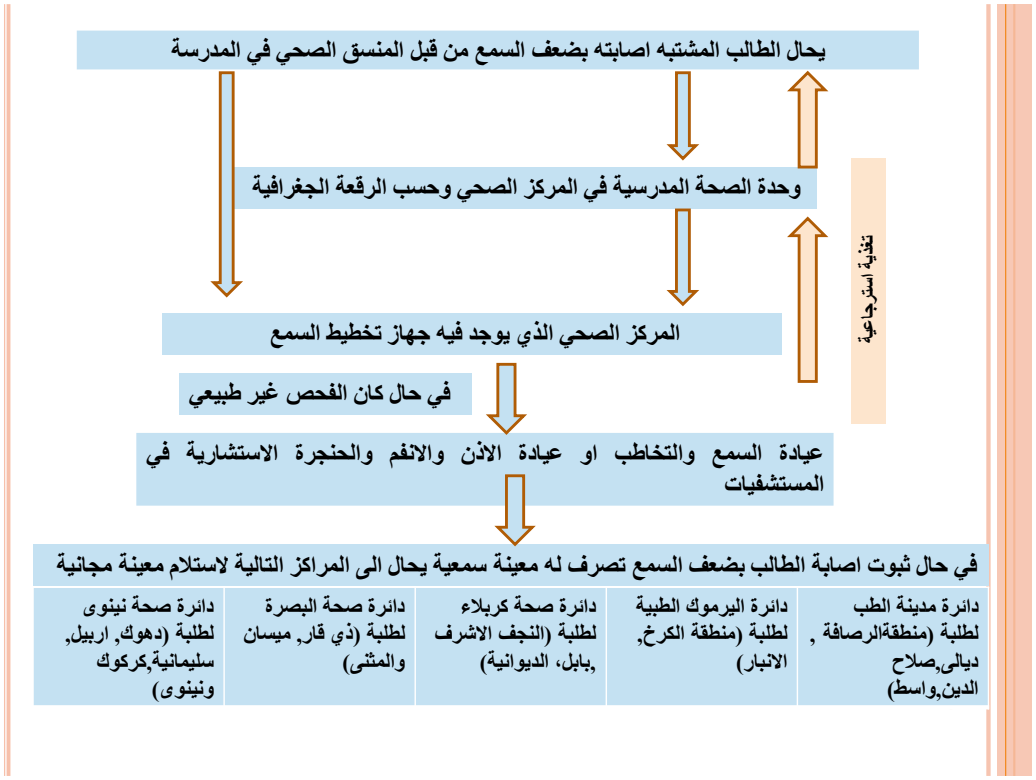
The School Health section in Health directorates, in coordination with the Directorate of Education and all primary health care sectors in governorates, places the presence of audiograms, which in turn supervises all health centers and schools so that they may be referred to the health centers where the device is located for diagnosis and treatment under three copies of the form. A sealed referral by the public health section in hospitals to follow up the procedures and ensure the feedback of the health centers.
Students diagnosed with hearing impairment in health centers are referred to consulting clinics or hearing and communication centers in hospitals for:

- Fixing the diagnosis and determining the necessary treatment or determining the type of audio needed by the student from the distribution outlets of the aids.
- Referring the student with severe hearing impairment to the special education committee, as appropriate, and documenting it in the form and the relevant records to determine admission to regular classes or special education classes.

(Scheme of referral of suspected cases of hearing impairment of new students)



(Scheme of referral of students suspected of hearing impairment in school)



Hearing Tests at Primary Health Care Centers



PRE-TEST

Questions	True	False	Don't know
128 Hz tuning fork is the most widely used for hearing tests			
Positive Rinne Test means that the patient is having conductive hearing loss			
In Weber test, Unilateral sensori-neural hearing –loss patients will lateralize the sound to the better ear			
In Whisper test, the unit of hearing test in the examined ear is measured according to the distance where the patient can hear the examiner's whispering			
Dispersion test needs two people to test the hearing			

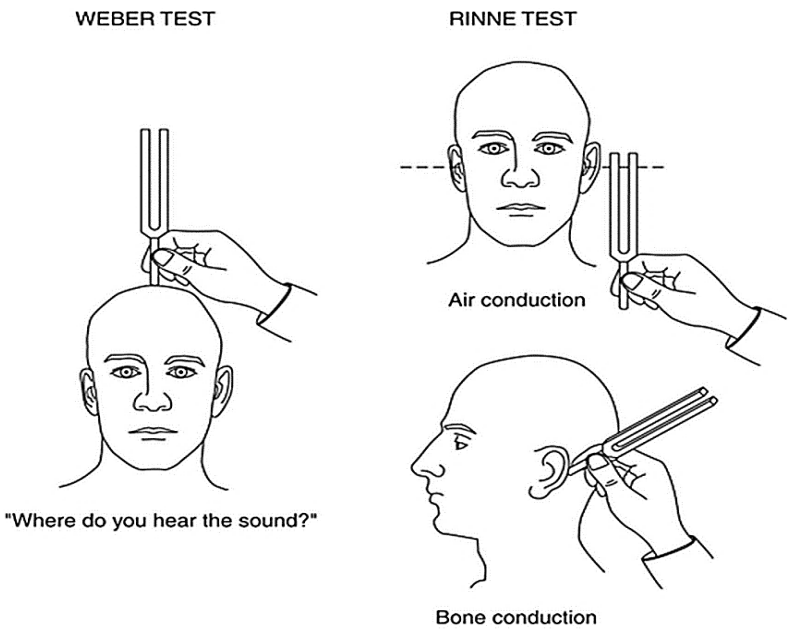
Whisper test:

- The patient sits on a chair facing the wall in a quiet room six meters long or more. The examiner stands behind the patient directly so that the patient cannot see the lips of the examiner.
- Cover one of the ears of the patient and the examiner whisper from behind him immediately, when the patient hear the examiner's whispering then the examiner move back one meter and repeat the whispering again and the examiner continues to move one meter to the back in case the patient was able to hear the whispering until he reaches a distance of six meter behind the patient which means that the patient has normal hearing and indicates that he has (6/6) according to the whisper test for the examined ear. The same test is repeated for the second ear.
- If the patient does not hear the examiner at a distance of less than six meters, for example, at the fourth meter, the examiner will proceed to the patient again at the third meter to make sure that the patient can hear at the third meter, and indicates that he has (3/6) according to the whisper test.
- The unit of hearing test in the examined ear is measured according to the distance where the patient can hear the examiner's whispering.

Tuning-fork tests:

Tuning-forks are basic hearing screening tools to other methods of hearing assessment; they may not support other audiological findings.

There are five tuning-fork test frequencies used clinically to test the hearing (as the Co fork at 128 Hz frequency is used for vibration testing rather than hearing testing): the C1-256Hz to the C5-4096Hz. The most useful fork is C2-512Hz because of the intermediate length of decay to allow enough time for minimizing the vibratory (rather than auditory)



stimulus when the fork is pressed against skin / bone, as happens with the lower frequencies forks.

1. Rinne test:

- Compares the air and bone conduction of the patient.
- The tuning fork is made to resonate and its base place in the mastoid process of the ear to be tested and the patient asked if he/ she can hear it (rather than feel it). A gentle but firm pressure is applied with the other examiners hand to the opposite side of the patients head to guarantee good contact of the forks base against the mastoid prominence.
- Then the tines of the fork are placed in line with the external auditory canal (approximately at 2.5-4 cm of distance).
- The patient is asked if he/ she can hear the sound louder in front or behind the ear, or the same in both.

Results:

- **Positive Rinne (**normal hearing or sensori-neural hearing loss): louder at the front or equally loud front or equally loud front and back. it means that air conduction is equal or better than bone conduction (AC ≥ BC).
- **Negative Rinne (**conductive hearing loss): Louder at the back. It means that bone conduction is better than air conduction (BC >AC).
- **A Rinne test** will detect a conductive defect in 50% of patients with an air-bone gap of 29 dB, and in 90% of patients with an air- bone gap of 40 dB.
- **Afalse-negative Rinne** test may appear when there is a severe unilateral sensori-neural hearing loss.BC is heard louder than AC, as the sound is transmitted through the skull to the contra lateral cochlea of the better hearing ear. Masking the opposite ear with a BARANY box will likely prevent this situation.

2. Weber test:

- Uses the bone conduction only & tests the lateralization of the sound.
- A resonating tuning fork is placed in the midline of the patient's skull (vertex) or forehead and the patient asked if he/she can hear it in the midline or towards one of the ears. Again, pressure may be applied to achieve good contact of the forks base

against the skull. If the patient cannot hear it at all, try placing the fork against the mandible or upper incisors (protecting the teeth with a surgical glove or similar).

- Normal patients or those with equal hearing loss will hear the forks sound centrally.
- Unilateral conductive hearing –loss patients will lateralize the sound to the poorer ear.
- Unilateral sensori-neural hearing –loss patients will lateralize the sound to the better ear.
- However, in some cases of unilateral hearing-loss or bilateral mixed hearing-loss the Weber test may not lateralize. It is important to conjugate Rinne and Weber tests results for more accurate diagnosis.

3. ABC test (Absolute bone conduction test):

- This test is so easy test depend on comparison between the patient & the examiner (who has normal hearing).
- This test done by using tuning fork applied on the mastoid bone of the patient & when the tone of the fork disappears, the fork removed from the patient to the mastoid bone of the examiner.

Result:

1. If the examiner hears the tone of the tuning fork so the patient had conductive deafness.
2. If the examiner didn't hear the tone of the tuning fork so the patient had no conductive deafness.

<6 months old	6 months - 3 years old	3 - 7 years old	7 - 18 years old
Observe response to sounds	Do a Distraction test of hearing*	Do a Voice test*	Do a Whispered voice test*
Response (+): Check again during next visit **Response (-)**: Counsel and refer			**Response (-)**: Do Tunning Fork Tests, Counsel and refer

POST-TEST

Questions	True	False	Don't know
128 Hz tuning fork is the most widely used for hearing tests			
Positive Rinne Test means that the patient is having conductive hearing loss			
In Weber test, Unilateral sensori-neural hearing –loss patients will lateralize the sound to the better ear			
In Whisper test, the unit of hearing test in the examined ear is measured according to the distance where the patient can hear the examiner's whispering			
Dispersion test needs two people to test the hearing			

Common Ear Infections



PRE-TEST

Questions	True	False	Don't know
Approximately 10% of people will be affected by otitis externa at some point during their life			
Acute otitis media is most common in young adults			
Escherichia coli is the most common pathogen causing chronic suppurative otitis media			
Ear infections can cause pain in the ear, fever, and temporary hearing loss			
If a child has tender swelling behind the ear, he is classified as having mastoiditis			
Ear discharge for two weeks or more is a feature of chronic ear infection			

1) OTITIS EXTERNA:

Otitis externa is an acute or chronic inflammatory reaction of the skin of the external ear canal.
Approximately 10% of people will be affected by otitis externa at some point during their life.

Local risk factors
- Warm, humid climates.
- Swimming (either in sea or swimming pool water).
- Change in ear canal pH (absence of cerumen).
- Trauma (from hearing aid moulds or foreign bodies).



Systemic factors
- Underlying skin disorders: allergic dermatitis, psoriasis, seborrheic dermatitis, eczema.
- Diabetes mellitus.
- Immunosuppression.

❖ ACUTE DIFFUSE OTITIS EXTERNA (SWIMMER'S EAR)

The most common form of otitis externa. It is a diffuse infection of the external ear canal.

Aetiology
The organisms most frequently involved are: *Pseudomona aeruginosa*, *Staphilococcus aureus*, and *Bacilus proteus*. This can be confirmed by taking an ear swab before treatment is started.
Heat and humidity are the main causative local factors (swimmer's ear), but also minor trauma (scratch) to the ear canal.

Clinical features
- Pain is the principal symptom, especially on moving the pinna or the jaw.
- Skin of the ear canal is swollen, red (infiltrative form), itchy, and there may be abundant debri (desquamative form) deeper in the canal and the tympanic membrane intact, but inflamed.
- There may be discharge but mucopus suggests middle ear origin and look for an ear drum perforation.

Treatment
- Aural toilet ± insertion of ear wick (small compacted dressing that once introduced in the ear canal expands in contact with ear drops applied onto it).

- Antibiotic–steroid combination drops (preferably anti-pseudomonal).
- Oral antibiotic (if there is no response to initial topical therapy, or spreading cellulitis).
- If the underlying factor/s contributing to acute otitis externa is/are not treated appropriately, recurrence of the episode and even chronicity may develop.
- Recurrent episodes can also lead to stenosis of the external ear canal.

❖ ACUTE LOCALIZED OTITIS EXTERNA (FURUNCULOSIS)

Less common form of otitis externa.



Aetiology
Caused by a Gram-positive bacterial infection (*Staph. aureus*, usually) of a hair follicle of the external ear canal.

Clinical features

- Forms a small abscess (furuncle).
- Extremely painful, localized swelling of the cartilaginous (outer) portion of the ear canal.
- There is no spreading cellulitits.
- Painful pinna to touch.
- Conductive deafness if the ear canal is occluded by the furuncle.
- It may coexist with diffuse otitis externa with accumulation of debri due to occlusion of the external ear meatus.
- May have associated post-auricular swelling (retro-auricular lymphadenopathy), forwardly pushing the pinna, which may mimic a subperiosteal mastoid abscess, especially if the tympanic membrane is not possible to be assessed.

Treatment
Aural toilet, which may include: incision and drainage of the small abscess (with a 21 FG needle), and/or insertion of a wick (ribbon gauze or pope wick) soaked in:

- Antibiotic-steroid combination drops, or in 10% glycerine-ichthammol solution.
- Oral anti-stapholococcal antibiotic.
- Adequate analgesia (a warm pad may be soothing).

❖ OTOMYCOSIS

Aetiology
Resembles acute diffuse otitis externa but the causative agents are fungi: *Aspergillus* species (*niger, flavus, fumigatus*); *Candida albicans*; *Penicillium* and others.

Clinical features
- Itching is a prominent clinical symptom.
- On otoscopy, abundant debri can be visualized, sometimes forming colonies that resemble cotton wool, in the case of *Candida albicans*, or black spores, in the case of *Aspergillus niger*.
- However, in many cases the diagnosis is based on microbiology culture of ear swab, or in failure to treatment after a significant period (3–4 weeks) with antibiotic drops (or orally).

Treatment
- Aural toilet (± insertion of ear wick).
- Antifungal drops (at least for 4 weeks as spores are highly resistant).
- There is no role for systemic antifungals.

2) ACUTE OTITIS MEDIA (AOM)

- Acute supurative otitis media is a very frequent condition affecting almost everyone at some point during their lifetime, and especially children. The peak incidence is in the 1st 12 months of life (23%).
- Acute suppurative otitis media defines inflammation of the middle ear caused by an infective organism.
- Clinically there is a middle ear effusion with other associated signs andsymptoms, including otalgia.
- The term 'acute' indicated that this process is <3weeks of duration.
- If it occurs three or more times in a 3-month period, it is called recurrent AOM.
- Infection lasting more than 3 months becomes chronic suppurative otitis media (CSOM).

Risk factors
- Age <7 years.
- Slightly more common in males.
- Presence of upper respiratory infection.
- Bottle feeding.
- Daycare nursery.
- Anatomical abnormalities (cleft palate, craniofacial anomalies).
- Immunological deficiencies (Immunoglobulin deficiencies, AIDS, leukemia, immuno-suppressants).

Aetiology

- Virus (thought to be the commonest cause):

- Influenza A and B.
- Rhinovirus.
- Respiratory syncytial virus.
- Mumps.
- Parainfluenza.
- Adenovirus.

- Bacteria:

- Streptococcus pneumoniae.
- Hemophillus influenza.
- Moraxella (Branhamella) *catarrhalis*.

Signs and symptoms

❖ The initial signs and symptoms usually develop following an upper respiratory tract infection or barotraumas, and are related to Eustachian-tube obstruction:

- Pressure build-up in the ear.
- Fullness sensation.
- Hearing loss.

At this stage, on otoscopy, there is inflammation and dullness of the tympanic membrane with possibly visible effusion.

❖ If the process continues there will be frank suppuration in the middle ear:

-Otalgia.
- Pyrexia.
- On otoscopy the tympanic membrane will be seen bulging or there may be muco-purulent otorrhoea (through a perforation).



- Small children will be irritable:
  - ➢ Crying and pulling from the ear.
  - ➢ Vomiting and/or diarrhea.
  - ➢ Sleepy and lethargic.
  - ➢ Febrile convulsions can also occur.



❖ If perforation occurs, there is an improvement of the otalgia. Secondary otitis externa can supersede the initial middle-ear otorrhoea.

The ear canal edema can make it difficult to see the tympanic membrane and make the correct diagnosis.

In 5% of cases the condition will become complicated with persistence of, or with new signs and symptoms.

Management

- Analgesia and antipyretics.
- Take a swab for culture and microbiological studies and consider topical antibiotic (± steroid) drops.
- There is controversy in the use of oral antibiotics.
- If the pain persists and there is fever the antibiotic is mandatory.
- Using antibiotics lead to; reduction in analgesia during AOM, and a reduction of serious complications, including mortality.

Complications of acute otitis media

- Extracranial:
  - Tympanosclerosis.
  - Tympanic membrane perforation.
  - Hearing loss (conductive or sensori-neural).
  - Facial-nerve palsy.
  - Acute mastoiditis.
  - Petrositis.
  - Labyrinthitis.
  - Chronic suppurative otitis media (CSOM).
- Intracranial:
  - Meningitis.
  - Extradural abscess.
  - Subdural abscess.
  - Intracerebral abscess.
  - Lateral sinus thrombosis.
  - Otitic hydrocephalus.

CHRONIC SUPPURATIVE OTITIS MEDIA (CSOM) WITHOUT CHOLESTEATOMA:

- CSOM without cholesteatoma can be defined as the presence of chronic otorrhoea with a tympanic membrane perforation.
- Prevalence: 0.6% in the adult population.

Pathogens

- Common: *Pseudomona aeruginosa, Staphylococcus aureus, Proteus* species.
- Less commonly: *Escherichia coli*, *Streptococcus pneumoniae*, and *Bacteroids* species.

Signs and symptoms

- Tympanic membrane perforation (dry if 'inactive disease').
- Recurrent or chronic muco-purulent otorrhoea ('active disease').
- Hearing loss (usually conductive, but a sensori-neural component may coexist).



Complication signs

- Facial-nerve palsy,
- Vertigo,
- Intracranial event.

EAR INFECTIONS IN CHILDREN

What Is an Ear Infection?

An ear infection, or otitis media, is the most common cause of earaches. Although this condition is a frequent cause of infant distress and is often associated with children, are a common problem in children. Is most common in infants and young children, especially those between the ages of 6 months and 3 years. By the age of one year, most children will have had one or more middle ear infections. About 50 percent of infants have at least one ear infection by their first birthday

The infection in the middle ear (the space behind the eardrum where tiny bones pick up vibrations and pass them along to the inner ear) very often accompanies a common cold, the flu, or other types of respiratory infections. This is because the middle ear is connected to the upper respiratory tract by a tiny channel known as the Eustachian tube. Germs that are growing in the nose or sinus cavities can climb up the Eustachian tube and enter the middle ear to start growing.

Untreated, ear infections can lead to more serious complications, including mastoiditis (a rare inflammation of a bone adjacent to the ear), hearing loss, perforation of the eardrum, meningitis, facial nerve paralysis.

What Causes an Ear Infection?

The middle ear is a small space behind the ear drum that is supposed to be well ventilated by air that normally passes up from behind the nose, through the Eustachian tube, keeping the middle ear clean and dry. When there is not enough fresh air ventilating the middle ear, such as when the Eustachian tube is clogged or blocked, the area becomes damp, stagnant, and warm, a perfect breeding ground for germs.

In children and infants, the Eustachian tube is often too soft or immature and has a harder time staying open. Allergies, post nasal drainage, sinus infections, common cold viruses and adenoid problems can all interfere with the Eustachian tube's ability to let air pass into the middle ear.

Ear infections can cause pain in the ear, fever, and temporary hearing loss and general signs such as loss of appetite and irritability. Some children get better without specific antibiotic treatment but most young infants benefit from use of an antimicrobial agent.

Ear infections most often develop after a viral respiratory tract infection, such as a cold or the flu. These infections can cause swelling of the mucous membranes of the nose and

throat, and diminish normal host deafness such as clearance of bacteria from the nose, increasing the amount of bacteria in the nose

Viral respiratory tract infections can impair Eustachian tube function. Normal Eustachian tube function is important for maintaining normal pressure in the ear. Impaired Eustachian tube function changes the pressure in the middle ear (like when you are flying in an airplane).

Fluid (called an effusion) may form in the middle ear and bacteria and viruses follow, resulting in inflammation in the middle ear, the increased pressure causes the eardrum to bulge, leading to the typical symptoms of fever, pain, and fussiness in young children

Winter is high season for ear infections. They often follow a cold. Some factors that increase a child's risk for middle ear infections include:

- Crowded living conditions
- Attending day care
- Exposure to second hand smoke
- Close contact with siblings who have colds
- Having a cleft palate
- Allergies that cause congestion on a chronic basis
- Not being breast-fed
- Bottle-feeding while lying down
- Infant with bronchiolitis
- Pacifiers
- Barometric trauma is another risk factor for a middle ear infection.



*Barometric trauma: The pressure in the middle ear rises when the airplane you are travelling on descends or when you ascend while scuba diving. If the Eustachian tube is not open, the pressure in the middle ear cannot be equalized, and thus, may cause injury, which increases the risk of an acute ear infection.

CLASSIFICATIONS OF EAR PROBLEM:

There are four classifications for ear problem:

- MASTOIDITIS
- ACUTE EAR INFECTION
- CHRONIC EAR INFECTION
- NO EAR INFECTION

Here is the classification table for ear problem from the *ASSESS & CLASSIFY* chart.

. Tender swelling behind the ear.	MASTOIDITIS	→ *Give first dose of an appropriate antibiotic.* → *Give first dose of paracetamol for pain.* → *Treat to prevent low blood sugar.* → *Refer urgently to hospital.*
. agonizing ear pain OR . Pus is seen draining from the ear & discharge is reported for less than 14 days.	ACUTE EAR INFECTION	→ **Give appropriate antibiotic for 10 days.** → **Give paracetamol for pain.** → Dry the ear by wicking. → Advise mother when to return immediately. → Follow up in 5 days.
. Pus is seen draining from the ear and discharge is reported for 14 days or more.	CHRONIC EAR INFECTION	→ Dry the ear by wicking → Refer to ENT specialist.
• No agonizing ear pain AND • No pus seen draining from the ear.	NO EAR INFECTION	→ Refer to ENT specialist.

1) MASTOIDITIS

If a child has tender swelling behind the ear, classify the child as having MASTOIDITIS.

2) ACUTE EAR INFECTION

Ear Infection Symptoms:

Symptoms of an ear infection in adolescents and older children may include ear aching or pain and temporary hearing loss. These symptoms usually come on suddenly
In infants and young children, symptoms of an ear infection can include:

- Fever (temperature higher than 100.4ºF or 38ºC
- Pulling on the ear
- Fussiness or irritability
- Decreased activity
- Lack of appetite or difficulty eating
- Vomiting or diarrhea
- Draining fluid from the outer ear (called otorrhea)



If there is agonizing ear pain or, if you see pus draining from the ear and discharge has been present for less than two weeks, classify the child's illness as (ACUTE EAR INFECTION).

3) CHRONIC EAR INFECTION

If you see pus draining from the ear and discharge has been present for two weeks or more classify the child's illness as (CHRONIC EAR INFECTION).

4) NO EAR INFECTION

If there is no agonizing ear pain and no pus is seen draining from the ear but he has other ear problem as (itching or foreign body), the child's illness is classified as NO EAR INFECTION. The children need no additional treatment and refer to ENT specialist.



POST-TEST

Questions	True	False	Don't know
Approximately 10% of people will be affected by otitis externa at some point during their life			
Acute otitis media is most common in young adults			
Escherichia coli is the most common pathogen causing chronic suppurative otitis media			
Ear infections can cause pain in the ear, fever, and temporary hearing loss			
If a child has tender swelling behind the ear, he is classified as having mastoiditis			
Ear discharge for two weeks or more is a feature of chronic ear infection			

Ototoxic Medication



PRE-TEST

Questions	True	False	Don't know
Symptoms of ototoxicity include partial or profound hearing loss, vertigo, and tinnitus			
Antibiotics in the aminoglycoside class, such as gentamicin and tobramycin, may produce cochleotoxicity			
Ototoxic effects are also seen with quinine, pesticides, solvents, asphyxiants (such as carbon monoxide) and heavy metalssuch as mercury and lead			
When combining multiple ototoxicants, the risk of hearing loss becomes greater			

Ototoxicity is the property of being toxic to the ear (oto-), specifically the cochlea or auditory nerve and sometimes the vestibular system, for example, as a side effect of a drug. The effects of ototoxicity can be reversible and temporary, or irreversible and permanent. There are many well-known ototoxic drugs used in clinical situations, and they are prescribed, despite the risk of hearing disorders, to very serious health conditions. Ototoxic drugs include antibiotics such as gentamicin, loop diuretics such as furosemide and platinum-based chemotherapy agents such as cisplatin. A number of nonsteroidal anti-inflammatory drugs (NSAIDS) have also been shown to be ototoxic. This can result in sensorineural hearing loss, dysequilibrium, or both. Some environmental and occupational chemicals have also been shown to affect the auditory system and interact with noise.

Signs& symptoms:

Symptoms of ototoxicity include partial or profound hearing loss, vertigo, and tinnitus. The cochlea is primarily a hearing structure situated in the inner ear. It is the snail-shaped shell containing several nerve endings that makes hearing possible. Ototoxicity typically results when the inner ear is poisoned by medication that damages the cochlea, vestibule, semi-circular canals, or the auditory/ vestibulocochlear nerve. The damaged structure then produces the symptoms the patient presents with.

Ototoxicity in the cochlea may cause hearing loss of the high-frequency pitch ranges or complete deafness, or losses at points between. It may present with bilaterally symmetrical symptoms, or asymmetrically, with one ear developing the condition after the other or not at all. The time frames for progress of the disease vary greatly and symptoms of hearing loss may be temporary or permanent.

The vestibule and semi-circular canal are inner-ear components that comprise the vestibular system. Together they detect all directions of head movement. Two types of otolith organs are housed in the vestibule: the saccule, which points vertically and detects vertical acceleration, and the utricle, which points horizontally and detects horizontal acceleration. The otolith organs together sense the head's position with respect to gravity when the body is static; then the head's movement when it tilts; and pitch changes during any linear motion of the head. The saccule and utricle detect different motions, which information the brain receives and integrates to determine where the head is and how and where it is moving.

The semi-circular canals are three bony structures filled with fluid. As with the vestibule, the primary purpose of the canals is to detect movement. Each canal is oriented at right angles to the others, enabling detection of movement in any plane.

The posterior canal detects rolling motion, or motion about the X axis; the anterior canal detects pitch, or motion about the Y axis; the horizontal canal detects yaw motion, or motion

about the Z axis. When a medication is toxic in the vestibule or the semi-circular canals, the patient senses loss of balance or orientation rather than losses in hearing. Symptoms in these organs present as vertigo, difficulties walking in low light and darkness, disequilibrium, oscillopsia among others. Each of these problems is related to balance and the mind is confused with the direction of motion or lack of motion. Both the vestibule and semi-circular canals transmit information to the brain about movement; when these are poisoned, they are unable to function properly which results in miscommunication with the brain.
When the vestibule and/or semi-circular canals are affected by ototoxicity, the eye can also be affected. Nystagmus and oscillopsia are two conditions that overlap the vestibular and ocular systems. These symptoms cause the patient to have difficulties with seeing and processing images. The body subconsciously tries to compensate for the imbalance signals being sent to the brain by trying to obtain visual cues to support the information it is receiving. This results in that dizziness and "woozy" feeling patients use to describe conditions such as oscillopsia and vertigo.

Cranial nerve VIII: is the least affected component of the ear when ototoxicity arises, but if the nerve is affected, the damage is most often permanent. Symptoms present similar to those resulting from vestibular and cochlear damage, including tinnitus, ringing of the ears, difficulty walking, deafness, and balance and orientation issues.

OTOTOXIC AGENTS:

- **ANTIBIOTICS:**

Antibiotics in the aminoglycoside class, such as gentamicin and tobramycin, may produce cochleotoxicity.Macrolide antibiotics, including erythromycin, are associated with reversible ototoxic effects. The underlying mechanism of ototoxicity may be impairment of ion transport in the stria vascularis. Predisposing factors include renal impairment, hepatic impairment, and recent organ transplantation.

- **LOOP DIURETICS:**

Certain types of diuretics are associated with varying levels of risk for ototoxicity. Loop and thiazide diuretics carry this side effect. The loop diuretic furosemide is associated with ototoxicity, particularly when doses exceed 240 mg per hour. The related compound ethacrynic acid has a higher association with ototoxicity, therefore it is preferred only for patients with sulfa allergies.

❖ CHEMOTHERAPUTIC AGENTS:

Platinum-containing chemotherapeutic agents, including cisplatin and carboplatin, are associated with cochleotoxicity characterized by progressive, high-frequency hearing loss with or without tinnitus (ringing in the ears). The severity of cisplatin-induced ototoxicity is dependent upon the cumulative dose administered and the age of the patient, with young children being most susceptible.The vinca alkaloids, including vincristine, are also associated with reversible ototoxicity.

❖ OTHER MEDICINAL OTOTOXICDRUGS:

At high doses, quinine, aspirin and other salicylates may also cause high-pitch tinnitus and hearing loss in both ears, typically reversible upon discontinuation of the drug.

Ototoxicants in the environment and workplace

Ototoxic effects are also seen with quinine, pesticides, solvents, asphyxiants (such as carbon monoxide) and heavy metalssuch as mercury and lead. When combining multiple ototoxicants, the risk of hearing loss becomes greater. As these exposures are common, this hearing impairment can affects many occupations and industries.

Treatment:

No specific treatment may be available, but withdrawal of the ototoxic drug may be warranted when the consequences of doing so are less severe than those of the ototoxicity. Co-administration of anti-oxidants may limit the ototoxic effects.

Ototoxic monitoring during exposure is recommended by the American Academy of Audiology to allow for proper detection and possible prevention or rehabilitation of the hearing loss through a cochlear implant or hearing aid. Successful monitoring includes a baseline test before, or soon after, exposure to the ototoxicant. Follow-up testing is completed in increments after the first exposure, throughout the cessation of treatment. Shifts in hearing status are monitored and relayed to the prescribing physician to make treatment decisions.

It is difficult to distinguish between nerve damage and structural damage due to similarity of the symptoms. Diagnosis of ototoxicity typically results from ruling out all other possible sources of hearing loss and is often the catchall explanation for the symptoms. Treatment options vary depending on the patient and the diagnosis. Some patients experience only temporary symptoms that do not require drastic treatment while others can be treated with medication. Physical therapy may prove useful for regaining balance and walking abilities. Cochlear implants are sometimes an option to restore hearing. Such treatments are typically taken to comfort the patient, not to cure the disease or damage caused by ototoxicity. There is no cure or restoration capability if the damage becomes permanent.

POST-TEST

Questions	True	False	Don't know
Symptoms of ototoxicity include partial or profound hearing loss, vertigo, and tinnitus			
Antibiotics in the aminoglycoside class, such as gentamicin and tobramycin, may produce cochleotoxicity			
Ototoxic effects are also seen with quinine, pesticides, solvents, asphyxiants (such as carbon monoxide) and heavy metalssuch as mercury and lead			
When combining multiple ototoxicants, the risk of hearing loss becomes greater			

Noise Induced Hearing Loss



PRE-TEST

Questions	True	False	Don't know
Noise induced hearing loss always affects both ears			
Noise induced hearing loss can be the result of a one-time exposure to severe sound waves, or constant exposure to high sounds over a long period of time			
Listening to large size MP3 players through headphones or earphones, band playing, loud concerts and sports stadiums are examples of recreational noise			
The necessary legislation must be promulgated and applied firmly to control and address noise sources			

The World Health Organization estimates that 1.1 billion young people worldwide may be at risk of hearing loss due to unsafe listening practices. Nearly half of all adolescents and adults (aged 12 to 35) in middle-income countries and high-income countries are exposed to inappropriate and unsafe levels of use of personal hearing aids.

What is noise induced hearing loss?

Excessive exposure to noise, whether prolonged or repeated exposure to noise or once exposure to very severe sound, causes damage to the hearing system and leads to hearing loss, called **noise induced hearing loss (NIHL).**
Noise induced hearing loss can occur immediately or may take a long time to be noticeable. It can be temporary or permanent and can affect one ear or both ears.

Causes of hearing loss caused by noise (NIHL):

NIHL can be the result of a one-time exposure to severe sound waves, such as explosion, or constant exposure to high sounds over a long period of time, such as noise from a carpentry shop.

Noise sources can be classified into several categories:

1. **Noise of transportation and roads**: which is considered the first cause of environmental noise in some countries





2. **Social and recreational noise:** Which occur in the residential environment, and come at the top of noise types. And Includes recreational activities such listening to large size MP3 players through headphones or earphones, loud concerts and sports stadiums, yelling.



3. Factory noise and craft workshops: It is considered one of the most dangerous types of occupational noise and is the source of factories or workshops where they affect the workers in these places and the population living next to the industrial zones. The hearing senses of large factory workers are affected day by day and may lead to long-term deafness.

Factories and workshops are a major source of noise, such as, iron and steel factories, diesel engine testing, textile and glass industry, mechanical carpentry workshops, wood cutting and paper mills and printing presses. Construction hammers and concrete foundations of construction companies.





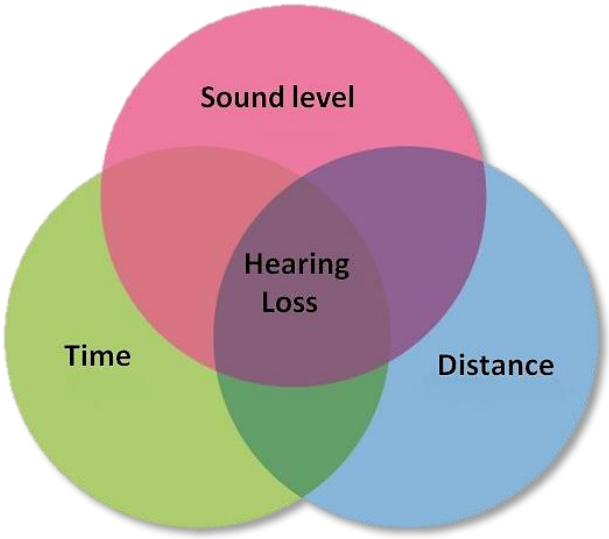
Sound is measured in units called **decibels**. Sounds less than 75 dB, even after prolonged exposure, are unlikely to cause hearing loss.

However, prolonged or repeated exposure to sounds at 85 dB or higher can cause hearing loss. The higher the sound, the less time it takes for hearing loss caused by noise.

Below are the average decibel estimates for some familiar voices:

- Humming of the refrigerator / 45 dB
- Normal conversation / 60 dB
- Noise from heavy traffic in the city / 85 dB
- Motorcycle / 95 dB
- MP3 player with maximum sound / 105 dB
- Siren / 120 dB
- Firecrackers and Firearms / 150 dB

The distance between you and the sound source and the length of time you are exposed to sound are important factors in hearing protection. A good rule is to avoid noise that is too high, too close, or long lasting



Key signs of hearing loss caused by noise:

When exposed to loud noise for a long period of time, it may slowly begin to lose hearing. Because noise damage is usually gradual, you may not notice it, or you may ignore the signs of hearing loss until it becomes clearer. The following are the most common signs and symptoms of hearing loss:

1. Often ask others to repeat their words.
2. Raise the sound of the TV or radio to a high level for others.
3. Has difficulty hearing through the phone.
4. Has difficulty to understand the conversations in crowded and noisy places.
5. Responds or responds incorrectly during conversations or discussions.

6. Has difficulty to hear the voice of women or children.
7. Feel that others are talking in unclear words (words incomprehensible).
8. Miss important information during meetings.
9. He avoids the social events he enjoyed in the past.
10. Continually advised by others to check his hearing.
11. Exposure to loud noise can also cause ringing in the ears or head. Tinnitus may calm down over time, but it may persist occasionally or sometimes throughout the person's life. Hearing loss and ringing can occur in one or both ears.

Prevention of noise induced hearing loss:

Noise induced hearing loss is the only type of hearing loss that can be completely prevented. If you understand noise hazards and how to exercise good hearing health, you can protect your hearing for life by knowing the factors that determine the impact of noise on your hearing:

- High intensity
- High frequency
- Continuous noise
- Personal readiness.

❖ **For children**:

Avoid using toys that make high sounds and do not use them near their ears

❖ **For youth:**

1. Wear earplugs or other protective devices when engaging in noise activity.
2. If you cannot reduce noise or protect yourself from it, stay away from it.
3. Use the head phone carefully and avoid high levels and long periods.

❖ **For workers:**

1. Personal safety equipment for workers such as ear protectors of all kinds when working in workshops and factories in which machines cause noise.
2. Design machines in a manner that reduces noise and maintenance.
3. Isolation of machines and devices that produce noise in soundproofed rooms such as generators
4. Replace old machines and machines that produce noise with another new less noise.

5. Conducting preliminary and periodic examinations of the workers to ensure the safety of hearing.
6. Reduction of daily working hours.
7. Change the work location of workers who have reduced hearing
8. Use of noise - proof materials in the walls and floors of plants with high noise.
9. Periodic maintenance of machines in the factories to reduce or eliminate noise, with increased control over industries and renewal of work permits.

❖ **For governments:**

1. The necessary legislation must be promulgated and applied firmly to control and address noise sources.
2. Health education through various media
3. Taking care of urban planning by exposing and landscaping the streets and increasing the area of parks and public parks within cities
4. Schools and hospitals should be away from sources of noise.
5. Airports should be away from cities and populated areas a distance of not less than30 km.
6. Make railways and highways as far away from residential areas as possible, and train wheels can be covered with rubber, as in underground trains.
7. Minimize the use of private transport routes and the trend towards public transport.
8. Reduce the use of alarm devices in cities.
9. The need to install sound insulation around the working buildings to reduce noise intensity
10. Limit the establishment of factories and power stations near the population centers.
11. Use noise prevention measures for factory and workshop workers.

POST-TEST

Questions	True	False	Don't know
Noise induced hearing loss always affects both ears			
Noise induced hearing loss can be the result of a one-time exposure to severe sound waves, or constant exposure to high sounds over a long period of time			
Listening to large size MP3 players through headphones or earphones, band playing, loud concerts and sports stadiums are examples of recreational noise			
The necessary legislation must be promulgated and applied firmly to control and address noise sources			

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